Number 2 — Standard form · Percentages · Percentage change

Calculator allowed. Give answers in standard form where the question is in standard form. Marks in brackets. ★ = harder question.

A · STANDARD FORM

1 Write in standard form: [4]

a $6\,800$ **b** 42 million **c** $0.000\,31$ **d** three eighths

2 Write as ordinary numbers: [4]

a 2.7×10^{4} **b** 9.05×10^{6} **c** 6×10^{-2} **d** 1.8×10^{-5}

3 Work out, giving each answer in standard form (3 s.f. where needed): [6]

a $(3.2 \times 10^{3}) \times (2.5 \times 10^{4})$ **b** $\dfrac{8.4 \times 10^{7}}{2.1 \times 10^{2}}$ **c** $6.5 \times 10^{5} + 6.5 \times 10^{4}$

d $(4 \times 10^{-3})^{2}$

4 Light travels at 3×10^{8} m/s. The Sun is 1.5×10^{11} m from the Earth. How long does sunlight take to reach us? Give your answer in minutes and seconds. [3]

B · PERCENTAGES

5 About 60% of the human body is water. How many kilograms of water are in a person of mass 65 kg? [2]

6 A laptop costs €840 before VAT. VAT is 24%. [4]

a Work out the price including VAT.

b In a sale, the price including VAT is reduced by 15%. Work out the sale price.

7 **a** A town's population grew from 2.4×10^5 in 2010 to 2.64×10^5 in 2020. Find the percentage increase. [4]

b The Earth is 1.496×10^8 km from the Sun and Mars is 2.279×10^8 km from the Sun. Write the Earth's distance as a percentage of the distance of Mars, to 3 s.f.

8 The length of a rectangle is increased by 20% and its width is decreased by 20%. Find the percentage change in its area. [2]

9 A marathon record of 2 hours 20 minutes was improved by 2.5%. Find the new record time in hours, minutes and seconds. [3]

Total: 32 marks

Check: in standard form the first number is between 1 and 10 · same power of 10 before adding · percentage change is measured against the *original* · a multiplier below 1 means a decrease.